Pathology of the Thyroid & Parathyroid

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Block: Endocrine/Urology/Reproduction

Thread: Pathology

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Textbooks

Robbins and Cotran: Pathologic Basis of Disease, 8th ed.—Chapter 24: The Endocrine System

Rubin's Pathology: Mechanisms of Human Disease, 8th ed.—Chapter 27: The Endocrine System

Learning Objectives

By exam time, you should be able to do the following:

1. Identify the main causes of hypo- and hyperthyroidism and their signs/symptoms
2. Describe and identify the histopathology of thyroid follicular nodular disease, Graves disease and Hashimoto thyroiditis
3. Explain the pathogenetic mechanisms underlying Graves disease and Hashimoto thyroiditis
4. Compare and contrast follicular adenoma, follicular carcinoma, papillary carcinoma, and medullary carcinoma
5. Distinguish between primary and secondary hyperparathyroidism based on clinical presentation, laboratory testing, and gross and microscopic pathological findings.
6. Compare and contrast the histopathology of parathyroid hyperplasia and parathyroid adenomas.

Study Questions

1. Explain how Hashimoto thyroiditis and Graves disease share a similar pathogenetic mechanism but lie at opposite ends of a physiologic spectrum. What is the mechanism behind each disease?

2. Identify the most common genetic alterations that occur in the following thyroid neoplasms: papillary thyroid carcinoma, follicular carcinoma, medullary thyroid carcinoma.

3. How would you distinguish primary hyperparathyroidism versus secondary hyperparathyroidism histologically? What is the most common cause of each?

Notes

A. Normal thyroid anatomy and function

1. Gross anatomy
    a. Two lateral lobes connected by a central isthmus ± pyramidal lobe (40%)
    b. Invested by a thin fibrous capsule containing numerous small vessels
2. Microscopic anatomy
    a. Follicle = functional unit of the thyroid
    b. Each follicle consists of a single layer of follicular cells (cuboidal epithelial cells) surrounding a central lumen filled with colloid
    c. Parafollicular (C) cells (larger cuboidal cells with abundant pale pink cytoplasm) are found scattered in between the follicles
3. Function
    a. Follicular cells secrete precursor glycoprotein called thyroglobulin (TG), which is stored in colloid
    b. In response to thyroid stimulating hormone (TSH) secreted by the anterior pituitary, follicular cells convert TG into T_3 and T_4 (active forms of thyroid hormone) by iodination of tyrosine residues, and then release them into perifollicular capillaries
    c. Thyroid hormones produce a wide array of physiological effects in virtually all of the body's organs and metabolic pathways
    d. Increased circulating levels of thyroid hormone have negative feedback on the pituitary (causing ↓TSH secretion) and hypothalamus (causing ↓TRH secretion)
    e. Parafollicular (C) cells are neuroendocrine cells that secrete calcitonin, which acts on the bones, kidneys and intestines to reduce blood levels of calcium, opposing the effects of parathyroid hormone

B. Hyperthyroidism

1. Definition
    a. Hypermetabolic state due to elevated circulating levels of thyroid hormone
2. Etiology
    a. Primary hyperthyroidism occurs due to an intrinsic abnormality of the thyroid, such as inflammation (thyroiditis), autoimmune disease (Graves disease), or a hyperfunctioning thyroid nodule that secretes excess thyroid hormone
    b. Secondary hyperthyroidism is caused by overstimulation of the thyroid by extrinsic sources, such as a TSH-secreting pituitary adenoma, which is not responsive to the normal regulatory control of feedback inhibition
3. Clinical features
    a. Most common signs and symptoms are nervousness/irritability, increased resting heart rate ± palpitations, elevated blood pressure, tremors, heat intolerance, weight loss with increase in appetite, mental disturbances, sleep disturbances, and menstrual abnormalities
4. Graves disease
    a. Most common cause of hyperthyroidism
    b. Autoimmune disorder caused by autoantibodies to the TSH receptor, which bind and *activate* the receptors and cause increased production of thyroid hormone
    c. Clinical triad = thyrotoxicosis, exophthalmos and pretibial myxedema
    d. Labs: ↑T_3 and T_4, ↓TSH (due to feedback inhibition)
    e. Pathology
        i. Gross: Diffusely enlarged gland
        ii. Microscopic: Diffuse hyperplasia of follicular epithelium with infoldings into lumen, columnar cell morphology and scalloping of the colloid

C. Hypothyroidism

1. Definition
    a. Condition in which the thyroid gland does not produce enough thyroid hormone
2. Etiology
    a. Primary hypothyroidism occurs due to an intrinsic abnormality in the thyroid causing defective synthesis of thyroid hormone, and the most common causes are iodine deficiency and Hashimoto thyroiditis
    b. Secondary hypothyroidism is due to hypothalamus or pituitary disease/damage, which leads to decreased secretion of TSH
3. Clinical features

   a. Slowed heart rate, cold intolerance, weight gain with decreased appetite, constipation, mental disturbances, hair loss, and menstrual abnormalities
4. Hashimoto thyroiditis
   a. Most common cause of hypothyroidism in geographic regions with sufficient iodine levels
   b. Autoimmune disease caused by autoantibodies to thyroid peroxidase and thyroglobulin, which lead to *destruction* of the thyroid tissue
   c. Labs: ↓T_3 and T_4, ↑TSH
   d. Risk factor for papillary thyroid carcinoma and B cell lymphoma of the thyroid
   e. Pathology
      i. Gross: Enlarged gland with pale cut surface
      ii. Microscopic: Chronic lymphocytic thyroiditis (lymphoplasmacytic inflammation ± germinal centers, follicular atrophy and oncocytic metaplasia)

D. Goiter

1. Definition
   a. Clinical term for enlargement of the thyroid gland (regardless of cause), which can be diffuse or nodular, and may produce excess thyroid hormone leading to hyperthyroidism ("toxic" goiter)
2. Causes
   a. Overstimulation of thyroid gland (e.g. Graves)
   b. Impaired synthesis of thyroid hormone (e.g. iodine deficiency, Hashimoto thyroiditis, drugs, goitrogens, inherited defects)
3. Multinodular goiter
   a. Etiology is multifactorial (genetic predisposition + environmental factors)
   b. More prevalent in iodine-deficient regions (endemic MNG)
   c. Impaired synthesis of thyroid hormone → low thyroid hormone levels → compensatory rise in TSH levels → hyperplasia of thyroid follicular cells and enlargement of the gland
   d. Multinodular appearance is caused by heterogeneity in follicular cells' responses to TSH, as well as repeated cycles of involution, degeneration and hyperplasia
   e. The increase in functional mass of the gland usually overcomes the thyroid hormone deficiency, so that the ultimate result is a euthyroid metabolic state
   f. May require surgical management if the enlarged gland causes compressive or obstructive symptoms

g. Pathology
    i. Gross: Enlarged gland with multiple nodules, some of which may show secondary changes such as cystic degeneration, hemorrhage or fibrosis
    ii. Microscopic: Thyroid follicular nodular disease (multiple variably sized nodules composed of admixture of small and large-sized follicles lined by benign follicular cells)

E. Thyroid neoplasia

1. Follicular adenoma (#1 benign thyroid neoplasm)
    a. Derived from follicular epithelium
    b. Usually nonfunctional, but may be "toxic" (produce excess thyroid hormone)
    c. Gross: well circumscribed and encapsulated
    d. Microscopic: cellular proliferation of microfollicles lined by tumor cells with small, round, hyperchromatic nuclei; NO capsular or vascular invasion (in contrast to follicular thyroid carcinoma)
2. Follicular thyroid carcinoma
    a. Microscopically identical to follicular adenoma except for the presence of invasion (either through the tumor capsule or into a blood vessel)
    b. Molecular alterations: *RAS* mutation, *PAX8/PPARG* fusion
3. Papillary thyroid carcinoma (#1 malignant thyroid neoplasm)
    a. Affects all ages and is more common in women
    b. May be multifocal
    c. Metastasizes through the lymphatics to cervical lymph nodes
    d. Associated with an excellent prognosis, even with nodal mets
    e. Pathology
        i. Papillary architecture (in classical type of PTC)
        ii. Psammoma bodies
        iii. Nuclear atypia: enlargement, elongation, overlapping, chromatin clearing, longitudinal grooves and intranuclear pseudoinclusions
    f. Molecular alterations: *BRAF* mutation, *RET/PTC* fusion
4. Medullary thyroid carcinoma
    a. Malignant neoplasm derived from parafollicular (C) cells
    b. Etiology/Molecular
        i. Approximately 70% of cases are sporadic and some are associated with somatic *RET* mutations
        ii. The remainder of cases are hereditary and associated with germline *RET* mutations (MEN2 syndromes)
    c. Pathology
        i. Variable morphologic appearance, can mimic other neoplasms
        ii. Typically nested/trabecular architecture (like other neuroendocrine tumors)

iii. Cellular morphology can be plasmacytoid, spindled or epithelioid
iv. Nuclear chromatin is granular ("salt and pepper") (like other neuroendocrine tumors)

F. Normal parathyroid anatomy and function

1. Gross anatomy
    a. 4 round glands (paired superior and inferior) on posterior aspect of thyroid
    b. Can be found in ectopic locations such as within the thyroid or thymus
    c. Each gland weighs 30-40 mg on average, but can weigh up to 70 mg
2. Histology
    a. Encapsulated organ composed of mixture of cell types and adipose tissue
    b. Normal cellularity is roughly 50%, but can range from 30-70%
    c. Chief cells have round nuclei and small amount of cytoplasm, but can sometimes accumulate more abundant clear cytoplasm due to increased glycogen content
    d. Oxyphil cells have round nuclei and abundant granular, eosinophilic (pink) cytoplasm (therefore are a type of oncocytic cell, analogous to oncocytic cells found in the thyroid)
3. Function
    a. Chief cells secrete parathyroid hormone (PTH)
        i. PTH has various actions on the bones, kidneys and intestines, which serve to increase calcium levels
        ii. Increase in blood calcium levels sends negative feedback to the parathyroid glands
        iii. PTH works in concert with calcitonin (secreted by thyroid) to maintain calcium homeostasis
    b. Function of oxyphil cells is unknown

G. Hyperparathyroidism

1. Etiology
    a. Primary hyperparathyroidism is the most common cause of hypercalcemia, and it is due to an intrinsic abnormality of the parathyroid gland(s) (most commonly an *adenoma*, less commonly, *hyperplasia* or *carcinoma*)
    b. Primary parathyroid hyperplasia and parathyroid adenoma may be associated with syndromes (e.g. MEN1, MEN2A)
    c. Secondary hyperparathyroidism occurs when chronic hypocalcemia causes compensatory *hyperplasia* of the parathyroid glands; this condition is most often seen in patients with chronic renal failure
2. Clinical features
    a. "Painful bones, renal stones, abdominal groans, psychic moans"

3. Parathyroid hyperplasia
    a. Increase in cellularity and, as a result, parenchymal mass, that typically affects at least 2, sometimes up to all 4, parathyroid glands
    b. Pathology: enlarged gland with increased cells (chief cells and/or oxyphil cells) and decreased fat
4. Parathyroid adenoma
    a. Benign neoplasm of chief cells and/or oxyphil cells
    b. Typically only one gland is affected, but "double adenomas" can occur
    c. Pathology: similar to parathyroid hyperplasia, except that oftentimes you can identify a rim of compressed normocellular parathyroid tissue at the periphery of the mass

Answers to Study Questions

1. Explain how Hashimoto thyroiditis and Graves Disease share a similar general pathogenetic mechanism but lie at opposite ends of a physiologic spectrum. What is the mechanism behind each disease?

Hashimoto thyroiditis (HT) and Graves Disease (GD) are both autoimmune conditions, with the former causing hypothyroidism and the latter causing hyperthyroidism.

In HT, patients have circulating antibodies to thyroid antigens, principally thyroglobulin and/or thyroid peroxidase. Through both antibody-dependent and antibody-independent mechanisms, thyroid follicular cells undergo apoptosis and are replaced by inflammatory cells and fibrosis, corresponding to the histologic appearance of a mononuclear inflammatory infiltrate with prominent germinal centers and oncocytic metaplasia (a metaplastic process of follicular epithelium to chronic injury). While there may be hyperthyroidism with the initial destruction of thyroid follicles and subsequent thyroid hormone release, hypothyroidism eventually results from replacement of the gland with inflammation and fibrosis.

In GD, the autoantibodies stimulate the TSH receptor on the follicular cells (hence the alternative term "thyroid stimulating immunoglobulin"), essentially mimicking pituitary TSH itself. However, in contrast to pituitary TSH, anti-TSH receptor antibodies of GD do not respond to feedback inhibition by peripheral T_3 and T_4. Hyperthyroidism results, with a low serum TSH (which should now make sense). Incidentally, the mechanism behind the infiltrative ophthalmopathy (exophthalmos) often appears confusing. There is experimental evidence in mice that orbital preadipocyte fibroblasts express the TSH receptor. After stimulation, it is possible that accompanying CD4+ T cells secrete cytokines that cause the fibroblasts to secrete extracellular matrix proteins into the retroorbital space, resulting in exophthalmos.

2. Identify the most common genetic alterations that occur in the following thyroid neoplasms: papillary thyroid carcinoma, follicular carcinoma, medullary thyroid carcinoma.

Papillary thyroid carcinoma: Most common =*BRAF* mutation (most commonly V600E) resulting in a gain of function in the MAP kinase pathway. Other = *RET/PTC* fusions (resulting from paracentric inversion or a translocation which results in a constitutively expressed receptor tyrosine kinase).

Follicular carcinoma: Most common = *RAS* gain of function mutations (leading to stimulation of the MAP kinase pathway). Others = *PIK3CA* mutations (leading to stimulation of PI3K signaling pathway); *PAX8-PPARG* fusions

Medullary thyroid carcinoma: *RET* mutations (but NOT *RET* fusions, as are seen in papillary thyroid cancer).

Notice how all three tumors arise from increased signaling in growth pathways.

3. How would you distinguish primary hyperparathyroidism versus secondary hyperparathyroidism histologically? What is the most common cause of each (primary vs. secondary hyperparathyroidism)?

Depending on the underlying cause of the primary hyperparathyroidism, you may not be able to distinguish primary and secondary hyperparathyroidism on histologic grounds alone. The most common cause of primary hyperparathyroidism is a parathyroid adenoma, which is a neoplasm. Histologically, therefore, one should see part of the gland that is hypercellular (the adenoma, >70% chief/oxyphil cells with little adipose tissue) and part of the gland that is normocellular (≤70% chief/oxyphil cells with abundant adipose tissue, the normal background parathyroid). In contrast, in parathyroid hyperplasia the entire gland should appear hypercellular. Often, all four glands are involved (though this need not be the case). Parathyroid hyperplasia can represent a primary parathyroid disease (causing primary hyperparathyroidism) or be a consequence of secondary hyperparathyroidism, most commonly caused by renal failure (whereby the hyperplasia is a physiologic adaptation to the hypocalcemia). The histologic features for both of these scenarios would be identical, so clinical correlation would be required to differentiate between the two.